UNIVERSITY OF MARYLAND

UMD DATA605 - Big Data Systems

11.1: Cloud Computing

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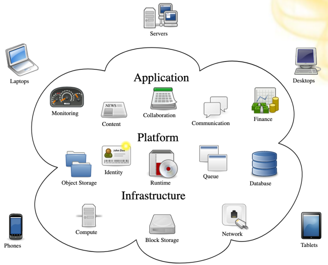
Cloud Computing

- Computing as "service" rather than "product"
  - Happens in the cloud: storage and computing
  - Edge devices (e.g., phones, laptops, tablets) interact with the cloud
- Advantages of cloud computing
  - Device agnostic: computation moves seamlessly between devices
  - On demand
  - Efficiency / scalability
    - Programming frameworks (e.g., Hadoop, Spark, Dask) enable scalability
  - Reliability
  - Cost: "pay-as-you-go" for computing resources
    - Cheaper than building own systems
    - Computing as a commodity (like electricity)



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Building vs Renting

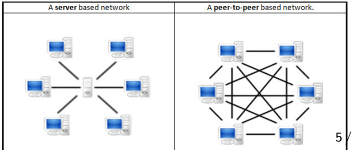
- **Building / buying infrastructure**
  - Require time and capital investments (Capex)
    - Especially at the beginning when there aren't revenues
    - A smooth cash flow (constant $/mo) is better than lumpy one (big one-time purchase)
  - Buy hardware (e.g., computers, storage, network)
    - How do you estimate the size of your hardware for today?
    - Even more difficult to estimate future demands
  - Update hardware when it becomes obsolete
  - Cost of owning hardware (Opex)
    - E.g., data center, electricity, cooling, handling faulty
  - Administering
    - Installing, updating, maintaining software stack

Building vs Renting

- **Cloud computing**
  - Pay for what you use
    - Low initial capital investment
  - Get your systems ready with a click on a web-site
  - No need to estimate a multi-year resource plan
  - Pick machines that suit your application and data requirements

Cloud Computing

- **Basic ideas of cloud computing around for a long time**
  - Mainframes + thin clients (1960s)
  - Personal computers (1980s)
  - Grid computing for supercomputers (1990s)
  - Peer-to-peer architecture (early 2000s)
  - Client-server model (Web 1.0 and Web 2.0)
  - Cloud computing (2010s)
- **It finally works**
- **Why now? A convergence of several key technologies over the last few years**
  - OS virtualization
  - Large data centers
  - Decreasing hardware costs
  - Big data frameworks

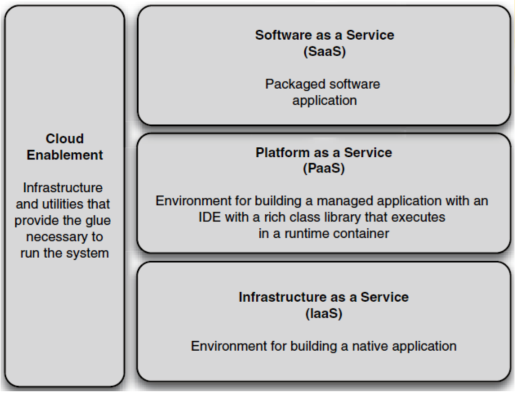


X-as-a-Service

Infrastructure-as-a-service (IaaS) - Cloud provides low-level resources - You install and maintain OS and applications - E.g., AWS EC2

Platform-as-a-service (PaaS) - Cloud provides OS and programming languages - You build application on top of it - E.g., Google App engine, managed Hadoop

Software-as-a-service (SaaS) - Cloud provides the application - You use it - No need to install anything on your machine - E.g., Dropbox, Salesforce, any app running in a browser

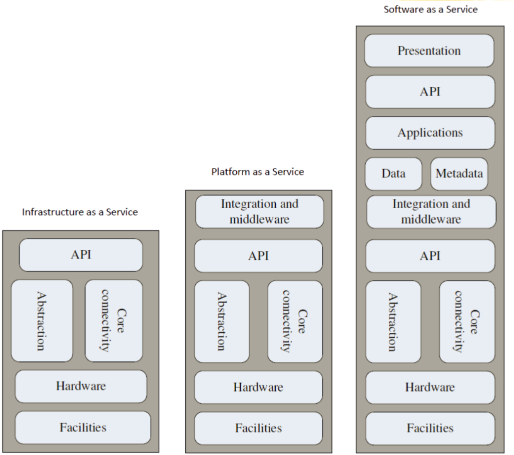


X-as-a-Service

... XaaS - Business model of "Anything-as-a-service" (2010s) - Desktop-as-a-service (e.g., AWS app) - Mobility-as-a-service (e.g., Uber) - Games-as-a-service (e.g., Google Stadia) - Storage-as-a-service (e.g., S3, Google Drive) - Marketing-as-a-service - Banking-as-a-service, . . .

Platform-as-a-Service

- **Problem: assembling your own software stack requires work**
- Install
- Configure
- Manage dependencies
- Incompatible versions
  - **Solution: get a pre-built software stack**
- Pre-installed OS
- Libraries
- Application software
- As a virtualization solution
  - E.g., VMware or Docker
  - The software stack comes as a large file with the image of the system
  - **Business model built**



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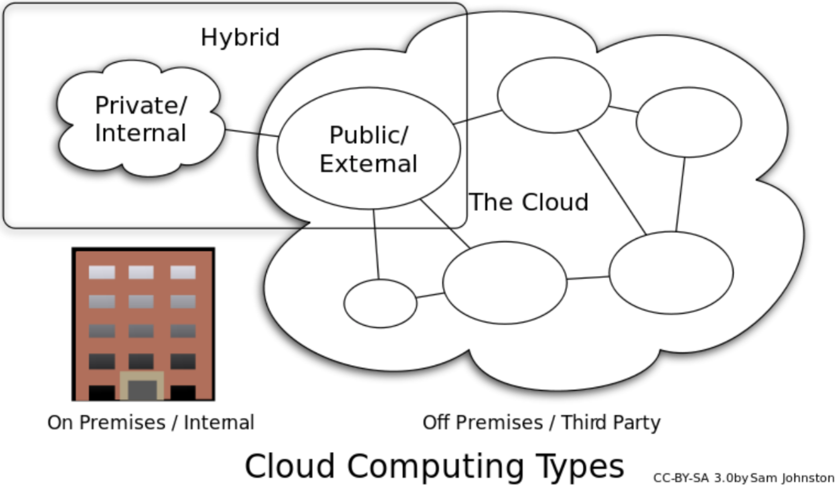
Different Cloud Computing Layers

From Dr. Mehmet Gunes, U. Nevada Reno

Cloud Computing Service Layers

From Dr. Mehmet Gunes, U. Nevada Reno

Cloud Types


Hybrid
Private/
Internal
Public/
External
The Cloud
On Premises / Internal
Off Premises / Third Party
Cloud Computing Types
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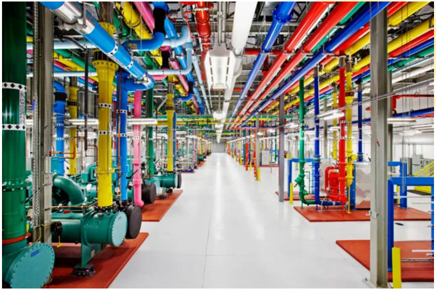
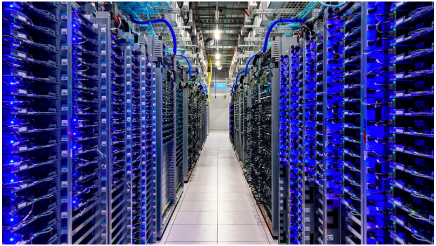
Outline

- Cloud Computing
- Technologies behind cloud computing
  - **Data centers**
  - Virtualization
  - Programming Frameworks
- Challenges and opportunities

Data Centers

- Data centers are key infrastructure piece that enables cloud computing
    - Every large company (e.g., AWS, Apple, Google, Facebook) is building data centers around the world
- Huge amount of work on deciding how to build / design them
- Research on how to save energy to power and cool data centers





Data Centers

- **Equipment cost**
  - E.g., computing, memory, storage, networking
  - A data center costs around 1B USD
  - Very expensive, but prices keep dropping
- **Powering / cooling cost**
  - Cost of running the equipment
  - Cost of cooling is quite high
    - Both have led to focus on "energy-efficient computing"
  - Appropriate placement of vents is a key issue
    - Thermal hotspots often appear and need to worked around

Data Center	Online	Buildings	SqFt (m)	Investment ($bn)
Dekalb, Illinois	2022	2	0.9	$0.8
Altoona, Iowa	2014	10	4.1	$2.0
Papillion (Sarpy), Nebraska	2019	8	3.6	$1.5
New Albany, Ohio	2020	5	2.5	$1.0
Huntsville, Alabama	2021	4	2.5	$1.0
Newton, Georgia	2023	5	2.5	$1.0
Forest City, North Carolina	2012	4	1.3	$0.8
Gallatin, Tennessee	2023	2	1.0	$0.8
Henrico, Virginia	2020	7	2.5	$1.0
Mesa, Arizona	Q4 2023	2	1.0	$0.8
Los Lunas, New Mexico	2019	6	2.8	$1.0
Fort Worth, Texas	2017	5	2.6	$1.5
Prineville, Oregon	2011	11	4.6	$2.0
Eagle Mountain, Utah	2021	5	2.4	$1.0
Odense, Denmark	2019	2	0.9	$1.6
Clonee, Ireland	2018	3	1.6	$0.4
Luleå, Sweden	2013	3	1.0	$1.0
Tanjong Kling, Singapore	2022	1	1.8	$1.0
Total		**85**	**39.6**	**$20.1**

Meta investment in 18 data centers

Data Centers

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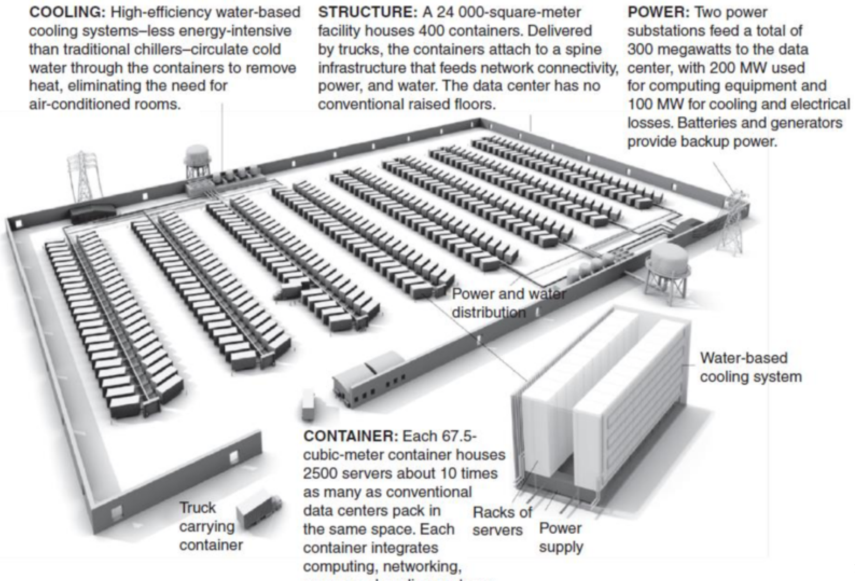
(Modular) Data Centers

From James Hamilton Presentation



Amazon Web Services

As of 2022, 28 geographical regions

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Amazon Web Services (EC2)

- The most widely used current solution to cloud computing
  - However alternatives are attractive depending on your needs
  - Current prices are very “low” and likely to remain that way due to competition
  - See https://aws.amazon.com/ec2/prici for current on-demand pricing

Viewing 564 of 564 available instances

Instance name	On-Demand hourly rate	vCPU	Memory	Storage	Network performance
a1.medium	$0.0255	1	2 GiB	EBS Only	Up to 10 Gigabit
a1.large	$0.051	2	4 GiB	EBS Only	Up to 10 Gigabit
a1.xlarge	$0.102	4	8 GiB	EBS Only	Up to 10 Gigabit
a1.2xlarge	$0.204	8	16 GiB	EBS Only	Up to 10 Gigabit
a1.4xlarge	$0.408	16	32 GiB	EBS Only	Up to 10 Gigabit
a1.metal	$0.408	16	32 GiB	EBS Only	Up to 10 Gigabit
t4g.nano	$0.0042	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t4g.micro	$0.0084	2	1 GiB	EBS Only	Up to 5 Gigabit
t4g.small	$0.0168	2	2 GiB	EBS Only	Up to 5 Gigabit
inf1.xlarge	$0.228	4	8 GiB	EBS Only	Up to 25 Gigabit
inf1.2xlarge	$0.362	8	16 GiB	EBS Only	Up to 25 Gigabit
inf1.6xlarge	$1.18	24	48 GiB	EBS Only	25 Gigabit
inf1.24xlarge	$4.721	96	192 GiB	EBS Only	100 Gigabit

Small Instance – default*

1.7 GB memory
1 EC2 Compute Unit (1 virtual core with 1 EC2 Compute Unit)
160 GB instance storage
32-bit platform
I/O Performance: Moderate
API name: m1.small

Large Instance

7.5 GB memory
4 EC2 Compute Units (2 virtual cores with 2 EC2 Compute Units each)
850 GB instance storage
64-bit platform
I/O Performance: High
API name: m1.large

Extra Large Instance

15 GB memory

Amazon S3

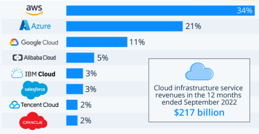
- Amazon storage services (Simple Storage Solution)
  - Pay for what you use

S3 Storage Types

	S3 Default	S3 RRS	S3 IA	Glacier
Durability	99.999999999%	99.99%	99.999999999%	99.999999
Availability	99.99%	99.99%	99.9%	99.99%
Extra Fees *	None	None	Retrieval	Retrieval
Real-Time Access?	Yes	Yes	Yes	No (mins/
Frequently Accessed?	Yes	Yes	No	No

Google App Engine

- Google Compute Engine (IaaS)
  - Directly compete with AWS EC2
- Google infrastructure (PaaS)
  - Run Docker container on Google resources
  - Managed services (e.g., SQL and NoSQL DBs)
- Google Docs (SaaS)
  - Share documents in the cloud
  - E.g., word processor, spreadsheet, presentations
- Google cloud computing market share
  - Built software infrastructure / data centers before Amazon
  - Invented many cloud technologies (e.g., Google File System, MapReduce, BigTable)
  - Market share is 3x smaller than AWS
  - Issues:
    - Developer / customer unfriendliness
    - (Often) lack of commitment (Killed by google)
    - No (or horrible) customer service



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- Cloud Computing
- Technologies behind cloud computing
  - Data centers
  - **Virtualization**
  - Programming frameworks
- Challenges and opportunities

Virtualization

- **Virtual machines have been around for a long time**
  - E.g., running Windows inside a Mac
  - Used to be very slow (e.g., QEMU)
- **Only recently (2000s) became efficient enough for cloud computing**
- **Basic idea of cloud computing**
  - Run virtual machines on your servers and sell time on them
  - E.g., Amazon EC2, Microsoft Azure, Google Cloud
- **Many advantages**
  - Security: virtual machines have almost impenetrable boundary
  - Multi-tenancy: can have multiple VMs on the same server
  - Efficiency: replace many underpowered machines with fewer high-powered machines

Virtualization

- **Consumer / desktop virtualization**
  - E.g., VMWare, Xen, VirtualBox
  - Run on top of host OS
  - Hypervisor / VM supports guest OS in virtual environment
- **Server virtualization**
  - Run hypervisor directly on top of hardware
  - Good for server farms, like in cloud computing
  - Amazon used Xen on RedHat machines
  - Now custom hardware AWS Nitro
  - Support Windows and Linux Virtual Machines

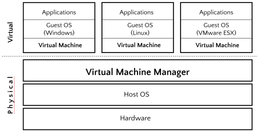

Figure 1: Consumer / desktop virtualization

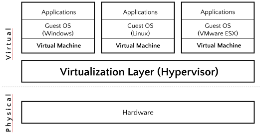

Figure 2: Server virtualization

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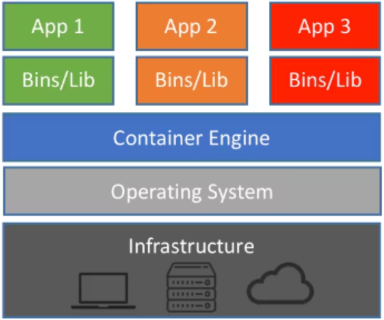
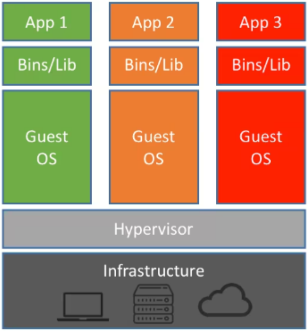
Virtualization

- Performance and tricky things to keep in mind
  - Hard to reason about performance (if you care)
  - Identical VMs may deliver somewhat different performance
    - E.g., multi-tenancy, different hardware
- "Bare-metal" compute to improve performance

Docker

- Enable true independence between application devs and IT ops
- Package all the dependencies
- Advantages
  - Containers are fast and portable
  - Reduce overhead of virtualization
  - Containers don't require full-blown OS
  - All containers run on a single host
  - Reduce OS licencing cost
  - Reduce overhead of OS maintenance





Machine Virtualization

Figure 3: alt_text

Outline

- Cloud Computing
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    - Data centers
    - Virtualization
    - **Programming frameworks**
- Challenges and opportunities

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Programming Frameworks

- **Programming frameworks emerged from efforts to "scale out" workloads**
  - Distribute work over large numbers of machines (thousands of machines)
- Parallelism has been around for a long time
  - Both in a single machine and as a cluster of computers
- Parallelism is very hard for programmers
  - Many things to keep track of:
    - How to parallelize application
    - How to distribute the data
    - How to handle failures
    - Debugging
    - Race conditions/Heisenbugs
    - ...
- **The difference is the user interface**
  - Google developed MapReduce and BigTable frameworks and started a new era
    - Hadoop, Spark
  - AWS services

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MapReduce Framework

- Provide a fairly restricted, but still powerful abstraction for programming distributed workloads
- Programmers only write functions called *map* and *reduce*
  - **map** programs
    - inputs: a list of “records” (e.g., data605/lectures_source/images, genomes)
    - output: for each record, produce a set of **(key, value)** pairs
  - **reduce** programs
    - input: a list of **(key, [values])** grouped together from the mapper
    - output: whatever
  - Both can do arbitrary computations on the input data as long as the basic structure is followed
- Everything else (e.g., task scheduling, fault tolerance) is taken care of by the framework
- **Batch processing of data**
  - MapReduce and other analytics frameworks are great
- **Streaming data**
  - Very large-scale applications that need real-time access with few ms latencies
  - Trade-off between “consistency” and “performance”

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Other Programming Frameworks

- Many different programming frameworks suitable for different applications
  - Address limitations of MapReduce-based platforms
- **High-performance Computing (HPC) Systems**
  - Cluster of supercomputers (instead of cheap computers)
  - E.g., GridRPC, MPI
  - More expressive and complex, but a lot more efficient
- **Spark**
  - Based on Resilient Distributed Data (RDD)
  - All in-memory, much more efficient
  - Uses Scala, Python, Java for programming
- **Apache Hive**
  - SQL-like interface on top of Hadoop / HDFS, originally from Facebook
- **Apache HBase**
  - NoSQL column oriented DB
  - Random read/write very large tables (like SQL) on top of Hadoop / HDFS
  - Modeled after Google BigTable
- **Apache Storm, Spark Streaming**
  - Focused on handling real-time streaming data
- **Giraph, GraphLab, GraphX**
  - Graph processing systems

Outline

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- **Challenges and opportunities**

Advantages of Cloud Computing

- **Lower edge computer costs**
  - Since applications run in the cloud, desktop does not need processing power / memory / disk demanded by traditional desktop software
- **Improved performance of desktop**
  - Better performance / faster system boot for your desktop due to large programs hogging your computer's memory
- **Device independence**
  - You are no longer tethered to a single computer
  - Applications and documents follow you through the cloud
  - Move to another (maybe portable) device and your applications and documents are still available
- **Reduced software costs**
  - Instead of purchasing expensive software applications, you can get most of what you need for free(-ish)
    - Google Docs suite vs Microsoft
    - Most applications are cloud based today
  - Rent (monthly payments) instead of buying

Advantages of Cloud Computing

- **Instant software updates**
  - No longer faced with choosing between obsolete software and high upgrade costs
  - When the application is web-based, updates happen automatically
    - Available the next time you log into the cloud
    - You get the latest version without needing to pay for or download an upgrade
  - Docker or VMs can package software to run on any platform that supports virtualization
- **Improved document format compatibility**
  - Do not have to worry about the documents on your machine being compatible with other users' applications or OSes
  - Potentially no format incompatibilities when everyone is sharing documents in the (same) cloud

Advantages of Cloud Computing

- **"Unlimited" storage capacity**
  - Cloud computing offers virtually limitless storage
  - Your computer has ~1TB hard drive
  - Hundreds of PBs available in the cloud (elastic and on-demand)
- **Increased data reliability**
  - Hard disk can crash and destroy all your valuable data of your desktop computer
  - Few individual users backup their data on a regular basis
  - A computer crashing in the cloud does not affect the storage of your data
    - All your data is still out there in the cloud, still accessible
    - Cloud computing is a data-safe computing platform

Advantages of Cloud Computing

- **Universal document access**
  - Documents stay in the cloud
  - Documents are instantly available from wherever you are
  - As long as you have an Internet connection
- **Latest version availability**
  - When you edit a document at home, that edited version is what you see when you access the document at work
  - The cloud always hosts the latest version of your documents
  - Built-in revision control system
- **Easier collaboration**
  - Sharing documents leads directly to better collaboration
  - Multiple users can collaborate easily on documents and projects

Disadvantages of Cloud Computing

- **Using cloud computing means dependence on Big Tech**
  - Big Internet companies like AWS, Google, and Microsoft monopolise the market
  - Can possibly limit flexibility and innovation
  - Lots of competition in cloud computing
  - High barriers to break into the cloud computing market
- **Security is an issue**
  - Unclear how safe outsourced data is
  - Ownership of data is not always clear
- **Issues relating to policy and access**
  - If your data is stored abroad whose policy do you adhere to?
    - To deal with government restrictions (e.g., export restrictions), cloud providers ensure your data is stored within your country's borders
    - E.g., TikTok saga
  - What happens if the remote server goes down?
    - How will you then access files?
  - Cases of users being locked out of accounts and losing access to data

Disadvantages of Cloud Computing

- **Requires a constant Internet connection**
  - Without an Internet connection you cannot access anything, applications or your own documents
  - No Internet means no work: this could be a deal-breaker
    - Many applications offer off-line capabilities
- **Does not work well with low-speed / spotty connections**
  - Low-speed Internet connection makes cloud computing painful or impossible
  - Web-based applications require a lot of bandwidth to download
- **Web-interface can be slow**
  - Even with a fast connection, web-based applications are often slower than your desktop
  - Interface (not data) is sent back and forth from your computer to the computers in the cloud
  - Latency (more than throughput) matters

Disadvantages of Cloud Computing

- **Features might be limited**
  - Many web-based applications simply are not as full-featured as their desktop-based applications
  - For example, you can still do a lot more with Microsoft PowerPoint than with Google Slides
  - This situation is changing rapidly
- **Stored data might not be secure**
  - With cloud computing, all your data is stored on the cloud
  - How secure is the cloud?
    - Probably more secure than your laptop
  - Can unauthorized users gain access to your confidential data?
    - News are data leaks are constant
- **Stored data can be lost**
  - Data stored in the cloud is safe, replicated across multiple machines (theoretically)
  - Doing local backups might still be a good idea

Disadvantages of Cloud Computing

- **HPC systems**
  - Unclear if you can run compute-intensive HPC applications that use MPI or OpenMP
  - Need low-latency, high-bandwidth interconnect
  - Scheduling is important with this type of application
  - All nodes should be co-located to minimize communication latency
  - This has changed (e.g., MapReduce, AWS EC2) and is still changing
- **Interoperability**
  - Each cloud systems uses different protocols and different APIs
  - May not be possible to run applications across different cloud-based systems
  - Solution: indirection layer (Terraform, Ansible)